§4.1 电磁场的矢势和标势

- 用势描述电磁场

  从真空中的有源的麦克斯韦方程组出发

$$\nabla\cdot\vec{E}=\frac{\rho}{\epsilon_0}\qquad\nabla\cdot\vec{B}=0\qquad\nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t}\qquad\nabla\times\vec{B}=\mu_0\vec{j}+\mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$$

  在静电和静磁情形下,为简化方程引入了标量势ϕ和矢量势$\vec{A}$,其中ϕ是建立在$\vec{E}$为保守场(无旋)的前提下的,即$\nabla\times\vec{E}=0$.现在这个条件一般不再满足,因此,原来的ϕ的定义$\vec{E}=\nabla\phi$不再一般适用. $\vec{A}$是建立在无磁单极的前提下,即$\nabla\cdot\vec{B}=0$.现在这个条件仍成立,因此,我们可以保持原来的$\vec{A}$的定义$\vec{B}=\nabla\times\vec{A}$. 它给出的物理意义是: 在任一时刻,$\vec{A}$沿任一闭合回路的线积分(环量)等于该时刻通过该回路内的磁通量. $\oint d\vec{l}\cdot\vec{A}=\int d\vec{S}\cdot\vec{B}$. 结合$\vec{A}$的定义和麦克斯韦方程组

$$\nabla\times\vec{E}=-\frac{\partial}{\partial t}\nabla\times\vec{A}=-\nabla\times\frac{\partial\vec{A}}{\partial t}\qquad\rightarrow\qquad\nabla\times(\vec{E}+\frac{\partial\vec{A}}{\partial t})=0$$

  在一般情况下,电场不仅受电荷激发,而且受变化磁场的激发,后者对电场的贡献是有旋的, 整个电场必须扣掉这部分才是无旋的,即$\vec{E}+\frac{\partial\vec{A}}{\partial t}$为无旋场,对它可引入标量势$\phi$

$$\vec{E}+\frac{\partial\vec{A}}{\partial t}=-\nabla\phi\qquad\rightarrow\qquad\vec{E}=-\nabla\phi-\frac{\partial\vec{A}}{\partial t}$$

- 规范变换和规范不变性

$$\begin{cases}\vec{E}=-\nabla\phi-\frac{\partial\vec{A}}{\partial t}\\ \vec{B}=\nabla\times\vec{A}\end{cases}\qquad\vec{A},\phi\text{不唯一}\qquad\begin{cases}\vec{A}\rightarrow\vec{A}'=\vec{A}+\nabla\chi\\ \phi\rightarrow\phi'=\phi-\frac{\partial\chi}{\partial t}\end{cases}$$

  每一组$(\vec{A},\phi)$叫一种规范,不同规范对应同一组$\vec{E},\vec{B}$, $\vec{E},\vec{B}$是物理观察量.

  所有物理量和物理规律与特殊的规范选择无关—规范不变性! 在数学上一种不变性就对应一种对称性,与上面规范不变性对应的对称性叫U(1)规范对称性. 因此, **电磁相互作用具有U(1)规范对称性.**

  在实际计算中,为了能有确定的$\vec{A},\phi$,总是选择一定的条件来把$\vec{A},\phi$所具有的不确定的规范自由度限制住,这种条件叫规范固定条件,选择一种规范固定条件叫选一种规范.常用的规范有,

  库伦规范: $\nabla\cdot\vec{A}=0$

  洛伦兹规范: $\nabla\cdot\vec{A}+\frac{1}{c^2}\frac{\partial\phi}{\partial t}=0$

- 达朗伯(d'Alembert)方程

  $\vec{A},\phi$的引入等价于已经求解了四个麦克斯韦方程中的两个, 剩下的两个方程给出,

$$\begin{aligned}&\nabla\times\vec{B}=\mu_0\vec{j}+\mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}\quad\rightarrow\quad\nabla\times(\nabla\times\vec{A})=\mu_0\vec{j}+\mu_0\epsilon_0[-\nabla\frac{\partial\phi}{\partial t}-\frac{\partial^2\vec{A}}{\partial t^2}]\\&\nabla\cdot\vec{E}=\frac{\rho}{\epsilon_0}\qquad\qquad\rightarrow\quad-\nabla^2\phi-\frac{\partial}{\partial t}\nabla\cdot\vec{A}=\frac{\rho}{\epsilon_0}\end{aligned}$$

(1).库伦规范: $\nabla\cdot\vec{A}=0$

$$\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}=-\mu_0\vec{j}+\frac{1}{c^2}\frac{\partial}{\partial t}\nabla\phi=-\mu_0\vec{j}^* \qquad \nabla^2\phi=-\frac{\rho}{\epsilon_0}$$

(2).洛伦兹规范: $\nabla\cdot\vec{A}+\frac{1}{c^2}\frac{\partial\phi}{\partial t}=0$

$$\nabla^2\vec{A}-\frac{1}{c^2}\frac{\partial^2\vec{A}}{\partial t^2}=-\mu_0\vec{j} \qquad \nabla^2\phi-\frac{1}{c^2}\frac{\partial^2\phi}{\partial t^2}=-\frac{\rho}{\epsilon_0}$$

注意,当$c\to\infty$时,回到静电磁情形.

§4.2 推迟势

- 非齐次(有源)波动方程的球面波解

  我们的目的是求解下面有源波动方程

$$\nabla^2\phi(\vec{r},t)-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi(\vec{r},t)=-\frac{\rho(\vec{r},t)}{\epsilon_0}$$

电荷分布$\rho(\vec{r},t)$可以看成是无穷多点源的迭加, $\rho(\vec{r},t)=\int d\tau'\delta(\vec{r}-\vec{r}')\rho(\vec{r}',t)$. 设点源$\delta(\vec{r}-\vec{r}')\rho(\vec{r}',t)$产生的电势分布为$U(\vec{r},\vec{r}',t)$, 即

$$\nabla^2_{\vec{r}}U(\vec{r},\vec{r}',t)-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}U(\vec{r},\vec{r}',t)=-\frac{1}{\epsilon_0}\delta(\vec{r}-\vec{r}')\rho(\vec{r}',t)$$

则容易验证(迭加原理)

$$\phi(\vec{r},t)=\int d\tau'\ U(\vec{r},\vec{r}',t)$$

$U(\vec{r},\vec{r}',t)$实际上是$\phi(\vec{r},t)$对应的格林函数. 从物理直观上看,位于$\vec{r}'$点的点源在场点$\vec{r}$产生的电势,应该关于$\vec{r}'$是球对称的,即

$$U(\vec{r},\vec{r}',t)=U(R,\vec{r}',t) \qquad \vec{R}=\vec{r}-\vec{r}'=\sum_{i=1}^{3}X_i\vec{e}_i$$

因此

$$\begin{aligned}\nabla^2_{\vec{r}}U(R,\vec{r}',t) &= \sum_{i=1}^{3}\partial_i\partial_iU(R,\vec{r}',t)=\sum_{i=1}^{3}\partial_i[\frac{\partial U}{\partial R}\partial_iR]=\sum_{i=1}^{3}\partial_i[\frac{\partial U}{\partial R}\frac{X_i}{R}]\\ &= \sum_{i=1}^{3}\Big(\frac{X_i}{R}\frac{\partial^2U}{\partial R^2}\partial_iR+\frac{3}{R}\frac{\partial U}{\partial R}+\frac{\partial U}{\partial R}X_i\partial_i\partial_i\frac{1}{R}\Big)=\sum_{i=1}^{3}\Big(\frac{X_iX_i}{R^2}\frac{\partial^2U}{\partial R^2}+\frac{3}{R}\frac{\partial U}{\partial R}-\frac{X_iX_i}{R^3}\frac{\partial U}{\partial R}\Big)\\ &= \frac{\partial^2U}{\partial R^2}+\frac{2}{R}\frac{\partial U}{\partial R}=\frac{1}{R}[\frac{\partial^2U}{\partial R^2}R+2\frac{\partial U}{\partial R}]=\frac{1}{R}\frac{\partial^2(RU)}{\partial R^2}\end{aligned}$$

结合原来的波动方程,在$R\neq0$(即$\vec{r}\neq\vec{r}'$)处,方程化为:

$$\frac{\partial^2(UR)}{\partial R^2}-\frac{1}{c^2}\frac{\partial^2(UR)}{\partial t^2}=0$$

这是标准的一维波动方程,它有两个独立解,即

$$RU(R,\vec{r}',t)=f(t-\frac{R}{c},\vec{r}')+g(t+\frac{R}{c},\vec{r}')$$

其中,$f(x,\vec{r}'),g(x,\vec{r}')$是两个任意函数.

$$U_1(R,\vec{r}',t)\equiv\frac{1}{R}f(t-\frac{R}{c},\vec{r}') \qquad \text{叫推迟解}$$
$$U_2(R,\vec{r}',t)\equiv\frac{1}{R}g(t+\frac{R}{c},\vec{r}') \qquad \text{叫超前解}$$

以上解是在$R\neq 0$处解出的,在$R=0$处有奇异. 而原始方程在$R=0$处也有奇异,要求它们奇异性相同,可定出f和g的形式.

$$\begin{aligned}
&-\frac{1}{\epsilon_0}\delta(\vec{R})\rho(\vec{r}',t)=(\nabla^2_{\vec{r}}-\frac{1}{c^2}\frac{\partial^2}{\partial t^2})[\frac{1}{R}f(t-\frac{R}{c},\vec{r}')]\\
&=(\nabla^2_{\vec{r}}\frac{1}{R})f(t-\frac{R}{c},\vec{r}')+2\nabla_{\vec{r}'}\frac{1}{R}\cdot\nabla_{\vec{r}}f(t-\frac{R}{c},\vec{r}')+\frac{1}{R}(\nabla^2_{\vec{r}}-\frac{1}{c^2}\frac{\partial^2}{\partial t^2})f(t-\frac{R}{c},\vec{r}')\\
&=-4\pi\delta(\vec{R})f(t-\frac{R}{c},\vec{r}')+\frac{2}{c}\frac{\vec{R}}{R^3}\cdot\frac{\vec{R}}{R}\frac{\partial}{\partial t}f(t-\frac{R}{c},\vec{r}')+\frac{1}{R}(\frac{\partial^2}{\partial R^2}+2\frac{\partial}{\partial R}-\frac{1}{c^2}\partial^2\partial t^2)f(t-\frac{R}{c},\vec{r}')\\
&=-4\pi\delta(\vec{R})f(t-\frac{R}{c},\vec{r}')=-4\pi\delta(\vec{R})f(t)
\end{aligned}$$

我们得到

$$f(t,\vec{r}')=\frac{\rho(\vec{r}',t)}{4\pi\epsilon_0}$$

所以

$$U_1(R,\vec{r}',t)=\frac{\rho(\vec{r}',t-\frac{R}{c})}{4\pi\epsilon_0 R} \qquad U_2(R,\vec{r}',t)=\frac{\rho(\vec{r}',t+\frac{R}{c})}{4\pi\epsilon_0 R}$$

这组解的特点是:
(1).对给定时刻t,等值面为以$\vec{r}'$为球心的球面,$t-\frac{R}{c}=$常数$\rightarrow R=|\vec{r}-\vec{r}'|=$常数.
(2).等值面以速率c运动,等值面的值的大小按$\frac{1}{R}$变化, 对推迟解,等值面扩张,等值面的值缩小,对超前解,等值面收缩,等值面的值增大.

考虑$t_2>t_1$, 等值面由$\vec{r}_1$移到$\vec{r}_2$,

$$\text{推迟解}: R_2=|\vec{r}_2-\vec{r}'|>R_1=|\vec{r}_1-\vec{r}'| \qquad t_1-\frac{R_1}{c}=t_2-\frac{R_2}{c} \qquad \frac{R_2-R_1}{t_2-t_1}=c$$
$$\text{超前解}: R_2=|\vec{r}_2-\vec{r}'|<R_1=|\vec{r}_1-\vec{r}'| \qquad t_1+\frac{R_1}{c}=t_2+\frac{R_2}{c} \qquad \frac{R_2-R_1}{t_2-t_1}=-c$$

推迟解告诉我们源点产生的电磁场是以光速c呈球面向外扩散的.
超前解告诉我们源点产生的电磁场是以光速c呈球面向内会聚的.
加入初始条件:源在某时刻t_0出现,t_0之前无电磁场,则可把超前解去掉.

- 洛伦兹规范

  洛伦兹规范下

$$\phi(\vec{r},t)=\int d\tau'\frac{\rho(\vec{r}',t-\frac{R}{c})}{4\pi\epsilon_0 R} \qquad \vec{A}(\vec{r},t)=\int d\tau'\frac{\mu_0\vec{j}(\vec{r}',t-\frac{R}{c})}{4\pi R}$$

对比静电磁情形,结果中多了一个R/c项,它代表推迟效应. 这种效应在一些情况下是不重要的, 如静态源$\rho,\vec{j}$与t无关,或似稳源, 源随时间变化周期$T \gg R/c$,这时源对t的依赖很小,可将c看成是∞, 理论可按静电磁场处理.

另一个与静电磁场不同之处是,推迟效应造成$\phi,\vec{A}$在无穷远处不全按$1/r$衰减.

以下验证上面的解确实满足洛伦兹规范的规范固定条件.

$$\begin{aligned}
\nabla\cdot\vec{A} &= \frac{\mu_0}{4\pi}\nabla\cdot\int d\tau'\frac{\vec{j}(\vec{r}',t^*)}{R} = \frac{\mu_0}{4\pi}\int d\tau'\,\nabla\cdot\left[\frac{\vec{j}(\vec{r}',t^*)}{R}\right] = \frac{\mu_0}{4\pi}\int d\tau'\,\left[(\nabla\frac{1}{R})\cdot\vec{j}(\vec{r}',t^*)+\frac{1}{R}\nabla\cdot\vec{j}(\vec{r}',t^*)\right]\\
&= \frac{\mu_0}{4\pi}\int d\tau'\,\left[-(\nabla'\frac{1}{R})\cdot\vec{j}(\vec{r}',t^*)+\frac{1}{R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t^*}\cdot\nabla t^*\right]\\
&= \frac{\mu_0}{4\pi}\int d\tau'\,\left[-\nabla'\cdot[\frac{\vec{j}(\vec{r}',t^*)}{R}]+\frac{1}{R}\nabla'\cdot\vec{j}(\vec{r}',t^*)+\frac{1}{R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t^*}\cdot\nabla t^*\right]\\
&= \frac{\mu_0}{4\pi}\int d\tau'\,\left[\frac{1}{R}\nabla'\cdot\vec{j}(\vec{r}',t^*)\Big|_{t^*}+\frac{1}{R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t^*}\cdot(\nabla t^*+\nabla' t^*)\right]\\
&= \frac{\mu_0}{4\pi}\int d\tau'\,(-\frac{1}{R})\frac{\partial}{\partial t^*}\rho(\vec{r}',t^*) = \frac{\mu_0}{4\pi}\int d\tau'\,(-\frac{1}{R})\frac{\partial}{\partial t}\rho(\vec{r}',t^*)\\
&= -\frac{\partial}{\partial t}\frac{\mu_0}{4\pi}\int d\tau'\,\frac{\rho(\vec{r}',t^*)}{R} = -\mu_0\epsilon_0\frac{\partial\phi(\vec{r},t)}{\partial t}
\end{aligned}$$

- 库伦规范

  库伦规范下

$$\phi(\vec{r},t)=\int d\tau'\frac{\rho(\vec{r}',t)}{4\pi\epsilon_0 R}\qquad \vec{A}(\vec{r},t)=\int d\tau'\frac{\mu_0\vec{j}^*(\vec{r}',t-\frac{R}{c})}{4\pi R}\qquad \vec{j}^*(\vec{r},t)=\vec{j}(\vec{r},t)-\epsilon_0\nabla\frac{\partial}{\partial t}\phi(\vec{r},t)$$

  对一个纯电流场,$\rho(\vec{r},t)=0\to\phi(\vec{r},t)=0$这时结果与洛伦兹规范相同.

§4.3 辐射电磁场

- 一般性质

  讨论源位于一有限区域,在很远处产生的场的行为:

$$\phi(\vec{r},t)=\int d\tau'\frac{\rho(\vec{r}',t^*)}{4\pi\epsilon_0 R}\qquad\qquad \vec{A}(\vec{r},t)=\int d\tau'\frac{\mu_0\vec{j}(\vec{r}',t^*)}{4\pi R}$$

  其中$t^*=t-\frac{R}{c}$

$$\vec{B}=\nabla\times\vec{A}\qquad\qquad \vec{E}=-\nabla\phi-\frac{\partial\vec{A}}{\partial t}$$

  若不考虑$\rho,\vec{j}$中对时间t依赖中推迟效应对$\phi,\vec{A}$的贡献,则在无穷远处$\phi\sim\frac{1}{r},\vec{A}\sim\frac{1}{r}\to\vec{E}\sim\frac{1}{r},\vec{B}\sim\frac{1}{r}$, 这时,

$$\oint_\infty d\vec{S}\cdot\vec{P}=\oint_\infty d\vec{S}\cdot(\vec{E}\times\vec{H})=0$$

  即:在无穷远表面上流出的电磁能量为0.电磁能量是不能流到无穷远的. 若要保证电磁能量能够流到无穷远, 至少$\vec{E}$和$\vec{B}$都按$\frac{1}{r}$在无穷远处衰减. 我们把具有这种性质的场叫辐射场.本节主要研究这种场的性

质.由于一般来说辐射场与非辐射场是混在一起的,为讨论方便,我们在以下讨论中只保留辐射场, 即只保留$r\to\infty$时场强正比$\frac{1}{r}$项的贡献.例:

$$\vec{n}\equiv\frac{\vec{R}}{R}=\frac{\vec{r}}{r}+O(\frac{1}{r})\qquad\frac{\partial t^*}{\partial t}=1\qquad\nabla t^*=-\nabla\frac{R}{c}=-\frac{\vec{n}}{c}$$
$$\nabla\psi(t^*)=\nabla t^*\frac{\partial\psi}{\partial t}=-\frac{\vec{n}}{c}\frac{\partial\psi}{\partial t}$$
$$\nabla\cdot\vec{f}(t^*)=\nabla t^*\cdot\frac{\partial\vec{f}}{\partial t}=-\frac{\vec{n}}{c}\cdot\frac{\partial\vec{f}}{\partial t}$$
$$\nabla\times\vec{f}(t^*)=\nabla t^*\times\frac{\partial\vec{f}}{\partial t}=-\frac{\vec{n}}{c}\times\frac{\partial\vec{f}}{\partial t}$$

辐射场为

$$\begin{aligned}
\vec{B}_{辐} &= (\nabla\times\vec{A})\Big|_{\frac{1}{r}}=\int d\tau'\frac{\mu_0}{4\pi R}\nabla\times\vec{j}(\vec{r}',t^*)=\int d\tau'\frac{\mu_0}{4\pi cR}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\times\vec{n}\\
\vec{E}_{辐} &= (-\nabla\phi-\frac{\partial\vec{A}}{\partial t})\Big|_{\frac{1}{r}}=\int d\tau'\Big[-\frac{1}{4\pi R\epsilon_0}\nabla\rho(\vec{r}',t^*)-\frac{\mu_0}{4\pi R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\Big]\\
&= \int d\tau'\Big[-\frac{1}{4\pi\epsilon_0 cR}\frac{\partial\rho(\vec{r}',t^*)}{\partial t}\vec{n}-\frac{\mu_0}{4\pi R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\Big]\\
&= \frac{\partial\phi}{\partial t}\frac{\vec{n}}{c}-\int d\tau'\frac{\mu_0}{4\pi R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}=-c\vec{n}\nabla\cdot\vec{A}-\int d\tau'\frac{\mu_0}{4\pi R}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\\
&= \int d\tau'\frac{\mu_0}{4\pi R}\Big[-c\vec{n}\nabla\cdot\vec{j}(\vec{r}',t^*)-\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\Big]=\int d\tau'\frac{\mu_0}{4\pi R}\Big[\vec{n}\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\cdot\vec{n}-\frac{\partial\vec{j}(\vec{r}',t^*)}{\partial t}\Big]
\end{aligned}$$

它满足

$$\vec{n}\times\vec{E}=c\vec{B}\qquad\qquad c\vec{B}\times\vec{n}=\vec{E}$$

因此,对辐射场$\vec{n},\vec{E},\vec{B}$互相垂直,成右手关系,辐射场的传播是横波, $\vec{E},\vec{B}$同位相

$$W_e=\frac{1}{2}\epsilon_0E^2\qquad W_m=\frac{1}{2}\mu_0H^2=\frac{B^2}{2\mu_0}=\frac{E^2}{2c^2\mu_0}=\frac{1}{2}\epsilon_0E^2=W_e$$
$$\vec{S}=\vec{E}\times\vec{H}=c\frac{B^2}{\mu_0}\vec{n}=2W_mc\vec{n}=(W_e+W_m)c\vec{n}$$

对比$\vec{j}=\rho\vec{v}$,电磁能量以c沿$\vec{n}$方向离开源传播.在每一不同时刻,$\vec{E},\vec{B}$可能各不相同,但$\vec{E}\times\vec{B}$总保持$\vec{n}$方向.

辐射功率

$$I=\int_{球}d\vec{\sigma}\cdot\vec{S}=\int_{球}d\sigma\ S\vec{n}\cdot\vec{n}=\int_{球}d\sigma\ S=\int d\Omega\ Sr^2=\int d\Omega\ \frac{dI}{d\Omega}\qquad\frac{dI}{d\Omega}=Sr^2$$

- 多极展开

由于我们只讨论远处的辐射场,在计算中只保留到$\frac{1}{r}$项.利用

$$R=|\vec{r}-\vec{r}'|=(r^2+r'^2-2\vec{r}\cdot\vec{r}')^{\frac{1}{2}}=r(1-\vec{n}\cdot\frac{\vec{r}'}{r}+\cdots)=r-\vec{n}\cdot\vec{r}'+\cdots$$

矢量势的辐射场部分为

$$\begin{aligned}
\vec{A}(\vec{r},t)\Big|_{\text{辐}} &= \int d\tau'\ \frac{\mu_0}{4\pi R}\vec{j}(\vec{r'},t-\frac{R}{c}) = \int d\tau'\ \frac{\mu_0}{4\pi R}\vec{j}(\vec{r'},t-\frac{r}{c}+\frac{\vec{n}\cdot\vec{r}'}{c}-\cdots)\\
&= \frac{\mu_0}{4\pi r}\int d\tau'\ \Big[\sum_{k=0}^{\infty}\frac{1}{k!}\frac{\partial^k\vec{j}(\vec{r'},t-\frac{r}{c})}{\partial t^k}(\frac{\vec{n}\cdot\vec{r}'}{c}-\cdots)^k\Big] = \frac{\mu_0}{4\pi r}\int d\tau'\ \sum_{k=0}^{\infty}\frac{1}{k!}\frac{\partial^k\vec{j}(\vec{r'},t-\frac{r}{c})}{\partial t^k}(\frac{\vec{n}\cdot\vec{r}'}{c})^k\\
&= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\frac{\partial^k}{\partial t^k}\int d\tau'\ \vec{j}(\vec{r'},t-\frac{r}{c})x'_{i_1}\cdots x'_{i_k}\\
&= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})
\end{aligned}$$

其中

$$\vec{J}_{i_1\cdots i_k}(t) = \int d\tau'\vec{j}(\vec{r}',t)x'_{i_1}\cdots x'_{i_k}$$

所以

$$\begin{aligned}
\vec{B}(\vec{r},t)\Big|_{\text{辐}} &= (\nabla\times\vec{A})\Big|_{\frac{1}{r}} = \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}\nabla\times\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\\
&= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^k}(-\frac{\vec{n}}{c}\times\frac{\partial}{\partial t})\frac{\partial^k}{\partial t^k}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\\
&= \frac{\mu_0}{4\pi r}\sum_{k=0}^{\infty}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n} = \sum_{k=0}^{\infty}\vec{B}_k
\end{aligned}$$

其中

$$\vec{B}_k = \frac{\mu_0}{4\pi r}\frac{1}{k!}\sum_{i_1,\cdots,i_k=1}^{3}\frac{n_{i_1}\cdots n_{i_k}}{c^{k+1}}\frac{\partial^{k+1}}{\partial t^{k+1}}\vec{J}_{i_1\cdots i_k}(t-\frac{r}{c})\times\vec{n}$$

注意$\vec{B}_{k+1}$比$\vec{B}_k$多$\frac{1}{c}x'\frac{\partial}{\partial t}\sim\frac{l'\omega}{c}\sim\frac{l'}{\lambda}$, 要展开好必须$l'\ll\lambda$. 特别地,

$$\begin{aligned}
\vec{B}_0 &= \frac{\mu_0}{4\pi r}\frac{\partial}{\partial t}\vec{J}(t-\frac{r}{c})\times\vec{n} = \frac{\mu_0}{4\pi r}\frac{\partial}{\partial t}\int d\tau'\vec{j}(\vec{r}',t-\frac{r}{c})\times\vec{n} = \frac{\mu_0}{4\pi r}\sum_{i,j,k=1}^{3}\frac{\partial}{\partial t}\epsilon_{ijk}\int d\tau'\ j_i(\vec{r}',t-\frac{r}{c})n_j\vec{e}_k\\
&= \frac{\mu_0}{4\pi r}\sum_{i,j,k=1}^{3}\epsilon_{ijk}\frac{\partial}{\partial t}\int d\tau'\ \Big[\partial'_l[j_l(\vec{r}',t-\frac{r}{c})x'_i]-x'_i\partial'_l j_l(\vec{r}',t-\frac{r}{c})\Big]n_j\vec{e}_k\\
&= \frac{\mu_0}{4\pi r}\sum_{i,j,k=1}^{3}\epsilon_{ijk}\frac{\partial}{\partial t}\int d\tau'\ x'_i\frac{\partial\rho(\vec{r}',t-\frac{r}{c})}{\partial t}n_j\vec{e}_k = \frac{\mu_0}{4\pi r}\sum_{i,j,k=1}^{3}\epsilon_{ijk}\frac{\partial^2}{\partial t^2}q_i(t-\frac{r}{c})n_j\vec{e}_k = \frac{\mu_0}{4\pi r}\ddot{\vec{P}}(t-\frac{r}{c})\times\vec{n}
\end{aligned}$$

其中

$$\vec{P}(t) = \sum_{i=1}^{3}q_i(t)\vec{e}_i \qquad\qquad q_i(t) = \int d\tau'\ x'_i\rho(\vec{r}',t)$$

这是电偶极辐射. 而

$$\vec{B}_1 = \frac{\mu_0}{4\pi c^2 r}\sum_{i=1}^{3}n_i\frac{\partial^2}{\partial t^2}\vec{J}_i(t-\frac{r}{c})\times\vec{n} = \frac{\mu_0}{4\pi c^2 r}\sum_{i=1}^{3}\frac{\partial^2}{\partial t^2}\int d\tau'\ x'_i\vec{j}(\vec{r}',t-\frac{r}{c})\times\vec{n}$$

$$
\begin{aligned}
&= \frac{\mu_0}{4\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\ x'_i j_j(\vec{r}', t-\frac{r}{c}) n_k \vec{e}_l \\
&= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\ [x'_i j_j + x'_j j_i + x'_i j_j - x'_j j_i] n_k \vec{e}_l \\
&= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \frac{\partial^2}{\partial t^2}\epsilon_{jkl}\int d\tau'\ \Big[\sum_{m=1}^{3}[\partial'_m(j_m x'_i x'_j) - x'_i x'_j(\partial'_m j_m)] + \sum_{m,n,p=1}^{3}\epsilon_{ijp}\epsilon_{nmp} x'_n j_m\Big] n_k \vec{e}_l \\
&= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \frac{\partial^2}{\partial t^2}\int d\tau'\ \Big[\epsilon_{jkl} x'_i x'_j \frac{\partial}{\partial t}\rho(\vec{r}', t-\frac{r}{c}) + \sum_{m,n,p=1}^{3}\epsilon_{jkl}\epsilon_{ijp}\epsilon_{nmp} x'_n j_m(\vec{r}', t-\frac{r}{c})\Big] n_k \vec{e}_l \\
&= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \Big[\frac{\partial^3}{\partial t^3}\epsilon_{jkl} q_{ij}(t-\frac{r}{c}) + 2\frac{\partial^2}{\partial t^2}\sum_{p=1}^{3}\epsilon_{jkl}\epsilon_{ijp} m_p(t-\frac{r}{c})\Big] n_k \vec{e}_l \\
&= \frac{\mu_0}{8\pi c^2 r}\sum_{i,j,k,l=1}^{3} n_i \Big[\frac{\partial^3}{\partial t^3}\epsilon_{jkl}\frac{1}{3} Q_{ij}(t-\frac{r}{c}) + 2\frac{\partial^2}{\partial t^2}\sum_{p=1}^{3}\epsilon_{jkl}\epsilon_{ijp} m_p(t-\frac{r}{c})\Big] n_k \vec{e}_l \\
&= \frac{\mu_0}{4\pi c^2 r}\Big[\frac{1}{6}\vec{n}\cdot \overset{\cdots}{\overrightarrow{\overrightarrow{Q}}} \times\vec{n} + (\ddot{\vec{m}}\times\vec{n})\times\vec{n}\Big]
\end{aligned}
$$

其中

$$
\overrightarrow{\overrightarrow{Q}}\ (t-\frac{r}{c}) = \int d\tau'\ (3\vec{r}'\vec{r}' - \overrightarrow{\overrightarrow{I}}\ r'^2)\rho\vec{r}', t-\frac{r}{c}) \qquad \vec{m}(t-\frac{r}{c}) = \frac{1}{2}\int d\tau'\ \vec{r}'\times\vec{j}(\vec{r}', t-\frac{r}{c})
$$

这是电四极辐射和磁偶极辐射的迭加.注意电四极辐射和磁偶极辐射属于同一阶.

- 电偶极辐射

设一短天线由A,B两个小球组成,($\vec{l}$很短,由B指向A),A,B上各带相反电荷, 充电后在A B间的导线上产生高频电流

$$
J = \frac{dQ_A}{dt} = J_0\cos\omega t \qquad\qquad Q_A = \frac{J_0}{\omega}\sin\omega t
$$

这个体系在$l\to 0, \frac{J_0}{\omega}\to 0$, 但$l\frac{J_0}{\omega}$固定的情况下成为一个严格的电偶极子,这个偶极子的的偶极矩$\vec{P} = Q_A\vec{l} = \frac{J_0}{\omega}\vec{l}\sin\omega t$是随时间变化的.

$$
\begin{aligned}
&\vec{B}(\vec{r}, t) = \frac{\mu_0}{4\pi c r}\ddot{\vec{P}}(t-\frac{r}{c})\times\vec{n} = -\frac{\mu_0 J_0\omega}{4\pi c r}\vec{l}\sin\omega(t-\frac{r}{c})\times\vec{n} = \frac{\mu_0\vec{l}}{4\pi c r}\dot{J}(t-\frac{r}{c})\times\vec{n} \\
&\vec{S}(\vec{r}, t) = \frac{c}{\mu_0}B^2\vec{n} = \frac{\mu_0 l^2}{(4\pi)^2 c r^2}\dot{J}^2(t-\frac{r}{c})\sin^2\theta\vec{n} \\
&\frac{dI}{d\Omega} = \frac{\mu_0 l^2}{(4\pi)^2 c}\dot{J}^2(t-\frac{r}{c})\sin^2\theta
\end{aligned}
$$

总辐射功率为

$$
\begin{aligned}
I(t) &= \int d\Omega\frac{dI}{d\Omega} = \int_0^{2\pi} d\phi\int_0^{\pi} d\theta\sin\theta\frac{dI}{d\Omega} = 2\pi\int_0^{\pi} d\theta\ \frac{\mu_0 l^2}{(4\pi)^2 c}\dot{J}^2(t-\frac{r}{c})\sin^3\theta = \frac{\mu_0 l^2}{6\pi c}\dot{J}^2(t-\frac{r}{c}) \\
&= \frac{\mu_0 l^2\omega^2 J_0^2}{6\pi c}\sin^2\omega(t-\frac{r}{c})
\end{aligned}
$$

平均总辐射功率为

$$
\begin{aligned}
\bar{I} &= \lim_{T\to\infty}\frac{1}{T}\int_0^T dt\ I(t) = \lim_{T\to\infty}\frac{1}{T}\int_0^T dt\ \frac{\mu_0 l^2\omega^2 J_0^2}{6\pi c}\frac{1}{2}[1-\cos 2\omega(t-\frac{r}{c})] \\
&= \frac{\mu_0 l^2\omega^2 J_0^2}{12\pi c}\lim_{T\to\infty}\frac{1}{T}[t-\frac{1}{2\omega}\sin 2\omega(t-\frac{r}{c})]\Big|_0^T = \frac{\mu_0 l^2\omega^2 J_0^2}{12\pi c} = \frac{\mu_0 l^2\omega^2}{6\pi c R}(\frac{1}{2}J_0^2 R) = \frac{R_{\text{辐}}}{R}\ (\frac{1}{2}J_0^2 R)
\end{aligned}
$$

其中

$$R_{辐}=\frac{\mu_0 l^2\omega^2}{6\pi c}\qquad\qquad \frac{1}{2}J_0^2R:\quad \text{热功率}\qquad\qquad R:\quad \text{天线电阻}$$

注意,$\overline{I}\nearrow$,则须$R_{辐}\nearrow$,即$\omega\nearrow$,因而只有高频电磁波才能辐射较大的能量,才能辐射的远.远距离无线通讯必须靠短波.

- 电四极辐射

  电量为$q,q,-2q$的三个小球分别位于长为$2l$的短天线的两端和中间,$\vec{l}$为从天线中点指向一端的矢量,且$q=q_0\cos\omega t,\ q_0\to 0,l\to 0$但$q_0l^2$固定.

$$\begin{aligned}
&J=\dot{q}=-q_0\omega\sin\omega t=J_0\sin\omega t\\
&\vec{P}=\int d\tau'\vec{r}'\rho=q\vec{l}+q(-\vec{l})=0\\
&\vec{m}=\frac{1}{2}\int d\tau'\vec{r}'\times\vec{j}=\frac{1}{2}\int\vec{r}'\times Jd\vec{l}'=0\\
&\overrightarrow{\overrightarrow{Q}}=\int d\tau' 3\vec{r}'\vec{r}'\rho=3\overrightarrow{\vec{l}\vec{l}}q+3(-\vec{l})(-\vec{l})q=6q\overrightarrow{\vec{l}\vec{l}}
\end{aligned}$$

$$\begin{aligned}
&\vec{B}(\vec{r},t)=\frac{\mu_0}{6c^2r4\pi}(\vec{n}\cdot\dddot{\overrightarrow{\overrightarrow{Q}}}\times\vec{n}=\frac{\mu_0}{6c^2r4\pi}6\dddot{\vec{q}}\ \vec{n}\cdot\overrightarrow{\vec{l}\vec{l}}\times\vec{n}=\frac{l^2\mu_0q_0\omega^3}{4\pi c^2r}\sin\omega(t-\frac{r}{c})\cos\theta\sin\theta\\
&\vec{S}(\vec{r},t)=\frac{c}{\mu_0}B^2\vec{n}=\frac{\mu_0l^4q_0^2\omega^6}{c^3r^2(4\pi)^2}\sin^2\omega(t-\frac{r}{c})\sin^2\theta\cos^2\theta\vec{n}\\
&\frac{dI}{d\Omega}=\frac{\mu_0l^4q_0^2\omega^6}{c^3(4\pi)^2}\sin^2\theta\cos^2\theta\sin^2\omega(t-\frac{r}{c})\\
&I=\frac{\mu_0l^4q_0^2\omega^6}{c^3(4\pi)^2}\sin^2\omega(t-\frac{r}{c})2\pi\int_0^\pi d\theta\ \sin^3\theta\cos^2\theta=\frac{\mu_0l^4q_0^2\omega^6}{30c^3\pi}\sin^2\omega(t-\frac{r}{c})\\
&\overline{I}=\frac{\mu_0l^4q_0^2\omega^6}{60c^3\pi}\sin^2\omega(t-\frac{r}{c})=\frac{\mu_0l^4J_0^2\omega^2}{60c^3\pi}=\frac{R_{辐}}{R}\ (\frac{1}{2}J_0^2R)
\end{aligned}$$

  其中

$$R_{辐}=\frac{\mu_0l^4\omega^4}{30\pi c}$$

- 磁偶极辐射

  电流圈半径为a,$I=I_0\cos\omega t,\vec{m}=I\pi a^2\vec{e}_z$

$$\begin{aligned}
&\vec{B}(\vec{r},t)=\frac{\mu_0}{4\pi c^2r}\ddot{\vec{m}}\times\vec{n}=\frac{\omega^2\mu_0\pi a^2}{4\pi c^2r}I_0\cos\omega t^*\vec{e}_z\times\vec{n}\\
&\vec{S}(\vec{r},t)=\frac{c}{\mu_0}B^2\vec{n}=\frac{\omega^4\mu_0a^4I_0^2}{16c^3r^2}\cos^2\omega t^*(\vec{e}_z\times\vec{n})\cdot(\vec{e}_z\times\vec{n})\vec{n}=\frac{\omega^4\mu_0a^4I_0^2}{16c^3r^2}\cos^2\omega t^*\sin^2\theta\vec{n}\\
&\frac{dI}{d\Omega}=\frac{\omega^4\mu_0a^4I_0^2}{16c^3}\cos^2\omega t^*\sin^2\theta\\
&I(t)=\frac{\pi^3\omega^4\mu_0a^4I_0^2}{6c^3}\cos^2\omega t^*\\
&\overline{I}=\frac{\pi^3\omega^4\mu_0a^4I_0^2}{12c^3}=\frac{R_{辐}}{R}\ (\frac{1}{2}J_0^2R)
\end{aligned}$$

  其中

$$R_{辐}=\frac{\mu_0a^4\omega^4\pi^3}{6c^3}$$

§4.4 电磁波的衍射

电磁波作为电磁场的波动满足波的一切基本性质. 波动的一个基本性质是惠更斯原理, 它的陈述是:在波传播的波阵面上的每一点都可以看作是次级波源,每一个这样的小波源发出小的球面子波,这些球面子波的包面形成新的波阵面. 这是对波的传播的定性描述, 定量的就是基尔霍夫公式.它建立了波在任意闭合面上和闭合面内任一点的定量联系.

设u代表一种波动,它可以是$E_x, E_y, E_z, B_x, B_y, B_z$,满足方程,

$$\nabla^2 u(\vec{r},t) - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}u(\vec{r},t) = 0$$

取$\vec{R} = \vec{r}' - \vec{r}$,并考虑$\vec{r} \subset \tau$,则,

$$\begin{aligned}
u(\vec{r},t) &= \int d\tau'\ \delta(\vec{r}'-\vec{r})u(\vec{r}',t-\frac{R}{c}) = \int d\tau'(-\frac{1}{4\pi})(\nabla'^2\frac{1}{R})u(\vec{r},t-\frac{R}{c}) \\
&= \int d\tau'(-\frac{1}{4\pi})\Big\{\nabla'[(\nabla'\frac{1}{R})u(\vec{r},t-\frac{R}{c})] - (\nabla'\frac{1}{R})\nabla' u(\vec{r},t-\frac{R}{c})\Big\} \\
&= \int d\tau'(-\frac{1}{4\pi})\Big\{\nabla'[(\nabla'\frac{1}{R})u(\vec{r},t-\frac{R}{c}) - \frac{1}{R}\nabla' u(\vec{r},t-\frac{R}{c})] + \frac{1}{R}\nabla'^2 u(\vec{r},t-\frac{R}{c})\Big\}
\end{aligned}$$

利用

$$\nabla' t^* = -\frac{1}{c}\nabla' R = -\frac{\vec{R}}{cR}$$

注意

$$\begin{aligned}
\frac{1}{R}\nabla'^2 u(\vec{r}',t-\frac{R}{c}) &= \frac{1}{R}\nabla'[\nabla' u(\vec{r}',t^*)\Big|_{t^*} - \frac{\partial u(\vec{r}',t^*)}{\partial t^*}\frac{\vec{R}}{cR}] \\
&= \frac{1}{R}\nabla'[\nabla' u(\vec{r}',t^*)\Big|_{t^*}] - \frac{\vec{R}}{cR^2}\nabla'\frac{\partial u(\vec{r}',t^*)}{\partial t^*} - \frac{1}{cR}\frac{\partial u(\vec{r}',t^*)}{\partial t^*}\nabla'\frac{\vec{R}}{R} \\
&= \frac{1}{R}\nabla'^2 u(\vec{r}',t^*)\Big|_{t^*} - \frac{\vec{R}}{cR^2}\cdot\frac{\partial}{\partial t^*}\nabla' u(\vec{r}',t^*)\Big|_{t^*} - \frac{\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big|_{t^*} \\
&\quad + \frac{1}{c^2 R}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}',t) - \frac{\partial u(\vec{r}',t^*)}{\partial t^*}\frac{1}{cR}[\frac{3}{R} - \frac{1}{R}] \\
&= \frac{1}{R}[\nabla'^2 u(\vec{r}',t^*)\Big|_{t^*} + \frac{1}{c^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}',t^*)] - \frac{2\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big|_{t^*} - \frac{2}{cR^2}\frac{\partial u(\vec{r}',t)}{\partial t^*} \\
&= \frac{2}{Rc^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}',t^*) - \frac{2\vec{R}}{cR^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big|_{t^*} - \frac{2}{cR^2}\frac{\partial u(\vec{r}',t)}{\partial t^*} \\
&= \frac{2}{c}\Big\{\frac{\vec{R}}{cR}\cdot\frac{\vec{R}}{R^2}\frac{\partial^2}{\partial t^{*2}}u(\vec{r}',t^*) - \frac{\vec{R}}{R^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big|_{t^*} - (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}',t^*)}{\partial t^*}\Big\} \\
&= -\frac{2}{c}\Big\{\frac{\vec{R}}{R^2}\cdot[-\frac{\vec{R}}{cR}\frac{\partial}{\partial t^*} + \nabla'\Big|_{t^*}]\frac{\partial}{\partial t^*}u(\vec{r}',t^*) + (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}',t^*)}{\partial t^*}\Big\} \\
&= -\frac{2}{c}\Big\{\frac{\vec{R}}{R^2}\cdot\nabla'\frac{\partial}{\partial t^*}u(\vec{r}',t^*) + (\nabla'\cdot\frac{\vec{R}}{R^2})\frac{\partial u(\vec{r}',t^*)}{\partial t^*}\Big\} \\
&= -\frac{2}{c}\nabla'\cdot\Big\{\frac{\vec{R}}{R^2}\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big\}
\end{aligned}$$

因此

$$u(\vec{r},t) = (-\frac{1}{4\pi})\int_\tau d\tau'\ \nabla'\cdot\Big\{(\nabla'\frac{1}{R})u(\vec{r}',t-\frac{R}{c}) - \frac{1}{R}\nabla' u(\vec{r}',t-\frac{R}{c}) - \frac{2}{c}\frac{\vec{R}}{R^2}\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big\}$$

$$
\begin{aligned}
&= (-\frac{1}{4\pi})\int_\tau d\tau'\ \nabla'\cdot\Big\{-\frac{\vec{R}}{R^3}u(\vec{r}',t-\frac{R}{c})-\frac{1}{R}\nabla' u(\vec{r}',t^*)\Big|_{t^*}-\frac{\vec{R}}{cR^2}\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big\}\\
&= \frac{1}{4\pi}\int_\tau d\vec{S}'\ \cdot\Big\{\frac{1}{R}\nabla' u(\vec{r}',t^*)+\frac{\vec{R}}{R^3}u(\vec{r}',t^*)+\frac{\vec{R}}{cR^2}\frac{\partial}{\partial t^*}u(\vec{r}',t^*)\Big\}\Big|_{t^*=t-\frac{R}{c}}
\end{aligned}
$$

将上述基尔霍夫公式用于讨论衍射问题.考虑平面波$u(\vec{r},t)$ 射向开有小孔的无穷大平面S上.作如下基尔霍夫假设:

- 除孔内外,$u,\nabla u$在S面上处处为0.
- 孔内的$u,\nabla u$的值等于没有任何屏或障碍物时,入射波的值.

即,入射波为,

$$
u(\vec{r},t)=\begin{cases}0 & \text{孔外}\\ Ae^{i(\omega t-\frac{\omega z}{c})} & \text{孔内}\end{cases}
$$

用基尔霍夫公式,场点的波为

$$
\begin{aligned}
u(\vec{r},t) &= \frac{A}{4\pi}\int_{\text{孔}} d\vec{S}'\cdot\Big\{\frac{1}{R}e^{i\omega t^*}(-\frac{i\omega}{c}\vec{e}_z)+\frac{\vec{R}}{R^3}e^{i\omega t^*}+\frac{\vec{R}}{cR^2}i\omega e^{i\omega t^*}\Big\} \qquad d\vec{S}'=-dS'\vec{e}_z\\
&= -\frac{A}{4\pi}e^{i\omega t}\int_{\text{孔}} dS'\Big\{\frac{-i\omega}{cR}+\frac{-\cos\theta}{R^2}-\frac{i\omega}{cR}\cos\theta\Big\}e^{-i\frac{\omega R}{c}}\\
&= \frac{A}{4\pi}e^{i\omega t}\int_{\text{孔}} dS'\Big\{\frac{i\omega}{cR}(1+\cos\theta)+\frac{\cos\theta}{R^2}\Big\}e^{-i\frac{\omega R}{c}}
\end{aligned}
$$

当场点离屏较远时,积分中主要是第一项的贡献,这种衍射叫夫琅和费衍射.
当场点离屏较近时,积分中主要是第二项的贡献,这种衍射叫菲涅尔衍射.

对一个长宽分别为a,b的矩形小孔,

$$
\begin{aligned}
&-\vec{R}=\vec{r}-\vec{\rho} \qquad \vec{\rho}=x\vec{e}_x+y\vec{e}_y \qquad \rho^2=x^2+y^2\\
&\vec{r}\cdot\vec{e}_x=r\cos\theta_1 \qquad \vec{r}\cdot\vec{e}_y=r\cos\theta_2 \qquad \vec{r}\cdot\vec{e}_z=r\cos\theta_3\\
&R=(r^2+\rho^2-2\vec{r}\cdot\vec{\rho})^{\frac{1}{2}}=[r^2+\rho^2-2r(x\cos\theta_1+y\cos\theta_2)]^{\frac{1}{2}}\\
&\cos\theta=\frac{-\vec{R}\cdot\vec{e}_z}{R}=\frac{r\cos\theta_3}{R}=\frac{r\cos\theta_3}{[r^2+\rho^2-2r(x\cos\theta_1+y\cos\theta_2)]^{\frac{1}{2}}}\\
&\rho\ll r\\
&R=r[1+\frac{\rho^2}{r^2}-\frac{2}{r}(x\cos\theta_1+y\cos\theta_2)]^{\frac{1}{2}}=r-x\cos\theta_1-y\cos\theta_2+\cdots\\
&\cos\theta=\cos\theta_3[1+\frac{x}{r}\cos\theta_1+\frac{y}{r}\cos\theta_2+\cdots]
\end{aligned}
$$

$$
\begin{aligned}
u(\vec{r},t) &= \frac{A}{4\pi}e^{i\omega t}\int_{\text{孔}} dS'\frac{i\omega}{cr}(1+\cos\theta_3)e^{-i\frac{\omega}{c}[r-x\cos\theta_1-y\cos\theta_2]}\\
&= \frac{A}{4\pi}e^{i\omega t}\frac{i\omega}{cr}(1+\cos\theta_3)\int_{-a}^{a}dx\int_{-b}^{b}dy e^{i\frac{\omega}{c}\cos\theta_1 x+i\frac{\omega}{c}\cos\theta_2 y}e^{-i\frac{\omega}{c}r}\\
&= \frac{A}{4\pi}e^{i\omega t-i\frac{\omega}{c}r}\frac{i\omega}{cr}(1+\cos\theta_3)\frac{2i\sin(\frac{\omega a}{c}\cos\theta_1)}{\cos\theta_1}\frac{2i\sin(\frac{\omega b}{c}\cos\theta_2)}{\cos\theta_2}\\
&= \frac{A}{4\pi}e^{i(\omega t-\frac{\omega}{c}r)}\frac{4ic}{\omega r}(1+\cos\theta_3)\frac{\sin(\frac{\omega}{c}a\cos\theta_1)}{\cos\theta_1}\frac{\sin(\frac{\omega}{c}b\cos\theta_2)}{\cos\theta_2}
\end{aligned}
$$

可以看出,衍射波为球面波,振幅在$\frac{\omega}{c}a\cos\theta_1=m\pi,\frac{\omega}{c}b\cos\theta_2=n\pi,(m,n,1,2,3)$处为0(暗点).